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Pore-Scale Assessment of CO_2 Solubility and Capillarity Trapping

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Abstract

Carbon Capture and Storage (CCS) has emerged as a pivotal technology in mitigating the escalating concern over anthropogenic CO_2 emissions. This study undertakes a pore-scale investigation into the efficacy of CCS, emphasizing CO_2 capillary and solubility trapping mechanisms within geological formations. Utilizing dynamic pore-scale imaging via x-ray micro-computed tomography (micro-CT), we examine the behavior of gaseous CO_2 injected into water-saturated Idaho Gray sandstone under various flow rates. Our methodology involves a CT scan imaging through a flow cell, enabling precise control and observation of CO_2 behavior during injection and imbibition processes. The research delineates the influence of injection rates on capillary trapping, revealing an optimal medium flow rate that maximizes CO_2 retention within the pore network. Additionally, the study delves into the solubility dynamics of CO_2 in response to pressure fluctuations, simulating real-world conditions of subsurface CO_2 storage. Key findings suggest that injection rates play a critical role in trapping efficiency, with a 33.96% trapping efficiency observed at an intermediate rate of 0.05 ml/min. Moreover, the study captures the temporal evolution of CO_2 solubility and exsolution, providing valuable insights into the long-term stability of sequestered CO_2. By enhancing the understanding of CO_2 phase behavior under varying reservoir conditions, this research offers significant contributions to the optimization of CCS strategies, with implications for both environmental sustainability and energy resource management.

Keywords: CO_2 storage, capillary pressure, capillary trapping, solubility trapping, multiphase flow, x-ray micro-CT, dynamic pore-scale imaging.

Introduction

One of the oil and gas industry's main challenges is minimizing the environmental carbon footprint. Burning fossil fuels to produce energy is associated with the high emission of CO_2(Hamieh et al. 2022). As oil demand grows, CO_2 emission is also expected to increase (Friedlingstein et al. 2020; IEA, 2022). The environmental impact of CO_2 drives the search for solutions to reduce its emission. Many approaches to reducing emissions include searching for alternative energy sources, imposing new policies, and cycling and capturing (Ye et al. 2023). Despite significant strides in exploring alternative energy sources and establishing new procedures, there remains considerable work to be done in recycling and carbon capture (Oelkers and Cole 2008; Gibbins and Chalmers 2008; Leung, Caramanna, and Maroto-Valer 2014; Cuéllar-Franca and Azapagic 2015; IPCC et al. 2021).

Carbon capture and storage (CCS) was established to provide practical solutions for the CO_2 challenge (Bachu 2008; Hulme 2016; IPCC 2014; P. Kelemen et al. 2019; Michael et al. 2010; Orr 2009; Rogelj et al. 2016). CCS technologies aim to capture CO_2, transport it, and store it underground (Ali et al. 2022; Alqahtani et al. 2023; A. Omar et al. 2021).

Carbon sequestration covers all methods of injecting CO_2 into the geological formation, aiming for long-term storage (Bachu, 2008; Kelemen et al., 2019; Michael et al., 2010; Orr, 2009). $CO_{2,}$ when injected into the formation, can take different routes based on the geological and petrophysical properties of the formation. CO_2 can be mechanically trapped in the formation of impermeable cap rock or trapped by capillarity in the presence of water. It can also dissolve in water and produce carbonic acid, which reacts with the rock to produce some minerals in a process known as mineralization (Addassi, Hoteit, and Oelkers 2024; Belshaw et al. 2024; Berndsen et al. 2024; Berner and Berner 2012; Gíslason et al. 2018; Goldberg, Takahashi, and Slagle 2008; Gunnarsson et al. 2018; P. B. Kelemen et al. 2020; Matter, Takahashi, and Goldberg 2007; McGrail et al. 2006; Oelkers and Gislason 2023; Snæbjörnsdóttir et al. 2020; Wang et al. 2024). These mechanisms are categorized into four main areas:

- Structural trapping
- Capillary or residual trapping
- Solubility trapping
- Mineralization

The Intergovernmental Panel on Climate Change (IPCC) recognizes the four mechanisms and reported in their Special Report Carbon Dioxide Capture and Storage (IPCC, 2005).

Capillary trapping is a vital mechanism in CCS as it provides a rapid and effective means of immobilizing a significant fraction of the injected CO_2, helping to ensure its long-term storage underground (Addassi et al. 2016; Krevor et al. 2015; Omar et al. 2021). It is also an essential mechanism in enhanced oil recovery, where it can influence the amount of oil that can be extracted from a reservoir. Understanding and optimizing capillary trapping can thus have significant implications for both subsurface operations' sustainability and profitability (add references).

Capillary trapping occurs due to the interplay of interfacial forces between a non-wetting fluid (in this case, CO_2) and a wetting fluid (like water) within a porous medium such as a geological formation. To understand this process, let us consider a scenario where CO_2 is injected into a water-filled porous formation. Initially, the CO_2 will displace the water in the pore spaces, a process similar to oil drainage. This displacement is primarily driven by the injected CO_2 pressure, which pushes the resident water toward the pore outlets. However, once the pressure of the injected CO_2 decreases, or if the injection stops, the water in the formation starts to imbibe again due to capillary forces. During imbibition, the water re-enters the pore spaces and pushes CO_2 toward the center of the pores. Due to the interfacial tension between the CO_2 and water, along with the capillary forces acting within the porous medium, some of the CO_2 gets trapped within the center of the pores. This trapped CO_2 is often in the form of disconnected droplets or blobs, unable to move or be flushed out by further water flow.

CO_2 capillary trapping has been studied experimentally and numerically(Bandara, Tartakovsky, and Palmer 2011; Benson and Cole 2008; Hesse, Orr Jr., and Tchelepi 2009; Kumar et al. 2005; Krevor et al. 2015). For instance, Iglauer et al. (2011) used a high-pressure, high-temperature flow cell to prove the feasibility of the $scCO_2$ capillary trapping as a CCS mechanism using a CT scan. They quantified the size of the trapped $scCO_2$ clusters. Krevor et al. (2015) also used a CT scan to study capillary trapping and concluded that trapped CO_2 can reach up to 10-30% of the pore volume and be permanently stable. They also highlighted that it comprises 60-95% of the storage capacity. Garing et al. (2017b) studied, with the help of CT scan imaging, the trapping physics of $scCO_2$. They highlighted in their study that the trapped $scCO_2$ by imbibition reconnects and forms larger ganglia, and no buoyancy effect was observed in their experiment.

CO_2 solubility trapping is indeed another critical mechanism in carbon capture and storage (Addassi et al. 2022; Burton and Bryant 2009; Eke et al. 2011; Emami-Meybodi et al. 2015; Shariatipour, Mackay, and Pickup 2016; Sigfusson et al. 2015). As its name suggests, it involves the dissolution of CO_2 into the resident brine in the storage formation. CO_2's solubility in water, while not high, can be influenced by several factors, including pressure, temperature, and salinity. These factors are typically inherent to the geological formations used for CO_2 storage.

Under the elevated pressures often found in these formations, CO_2's solubility in water can increase significantly. This increased solubility can lead to a potential enhancement in the overall efficiency of CO_2 storage. It essentially allows for additional CO_2 to be stored as a dissolved phase within the formation's resident brine, beyond what can be stored as a free phase in the pore spaces.

However, it is important to note that solubility trapping is a dynamic process and is sensitive to changes in the reservoir conditions. Since the solubility of CO_2 is a function of pressure, any variations in pressure during injection periods or shut-in periods can affect the amount of CO_2 that remains dissolved in the brine.

In particular, a decrease in pressure can lead to some of the dissolved CO_2 being released from the fluid. If this released CO_2 reaches a certain saturation, it can form connected pathways within the pore network. This newly mobilized CO_2 phase can then potentially migrate within the formation and, under certain conditions, could pose a risk of leakage.

The implications of these dynamics are significant. For one, they emphasize the importance of careful reservoir management to maintain the conditions favorable for solubility trapping. They also highlight the need for rigorous monitoring and modeling of the reservoir conditions and the stored CO_2 to ensure that any potential risks are identified and mitigated in a timely manner.

Moreover, they underline the importance of a comprehensive approach to CO_2 storage that takes into account all the different trapping mechanisms – structural, residual (or capillary), solubility, and mineral trapping (Addassi et al. 2022; 2021; Oelkers et al. 2022). Each of these mechanisms can contribute to the effective and secure storage of CO_2, but they also each have their dependencies and potential risks. Understanding these mechanisms and their interactions is thus key to maximizing the potential of CCS as a tool for mitigating climate change.

In this research, our objective is to extend our understanding of CO_2 sequestration mechanisms, focusing specifically on the role of capillary forces in trapping CO_2. Considering the dominance of previous studies based on supercritical CO_2, we aim to fill a knowledge gap by examining the dynamics of gaseous CO_2 when injected into the subsurface. This research investigates the potential for capillary trapping of CO_2 in its gaseous state and concurrently explores its solubility in water. We also delve into the complexities of CO_2 exsolution dynamics from aqueous solutions due to variations in pressure. Such a process parallels the phenomena observed in subsurface reservoirs, specifically during the injection of CO_2-laden water into basalt formations.

The implications of this research are twofold. Firstly, by improving our understanding of how capillary forces can trap gaseous CO_2, we can enhance the efficiency and effectiveness of CO_2 sequestration strategies. Secondly, by examining the solubility of CO_2 in water under different conditions, we can gain insights to the behavior of CO_2 following injection into subsurface formations.

Material and Methods

Materials

The wetting phase used in this experiment is 5 wt% Cesium Chloride solution, prepared using deionized (DI) water produced from a Milli-Q water purification system. Cesium Chloride was used enhance contrast between the water, CO_2, and the solid grains during imaging. The Cesium Chloride was sourced from Alfa Aesar, while the CO_2 gas, with a purity greater than 99.995%, was obtained from AirLiquide. The **mini-core plugs** used in the experiments were prepared from Idaho Gray sandstone. These core plugs had a diameter of 6.35 mm and lengths ranging from 10 to 15 mm. The plugs were cut to the desired dimensions using a high-precision saw, and a ¼-inch drilling bit was employed to drill through the cut rock using a MetaRock Universal Turret Milling Machine (Figure 1). Following drilling, the roughness of the core surfaces was refined with a high-precision polishing machine to ensure smoothness. All mini-core plugs were thoroughly cleaned and dried prior to use in the experiments.

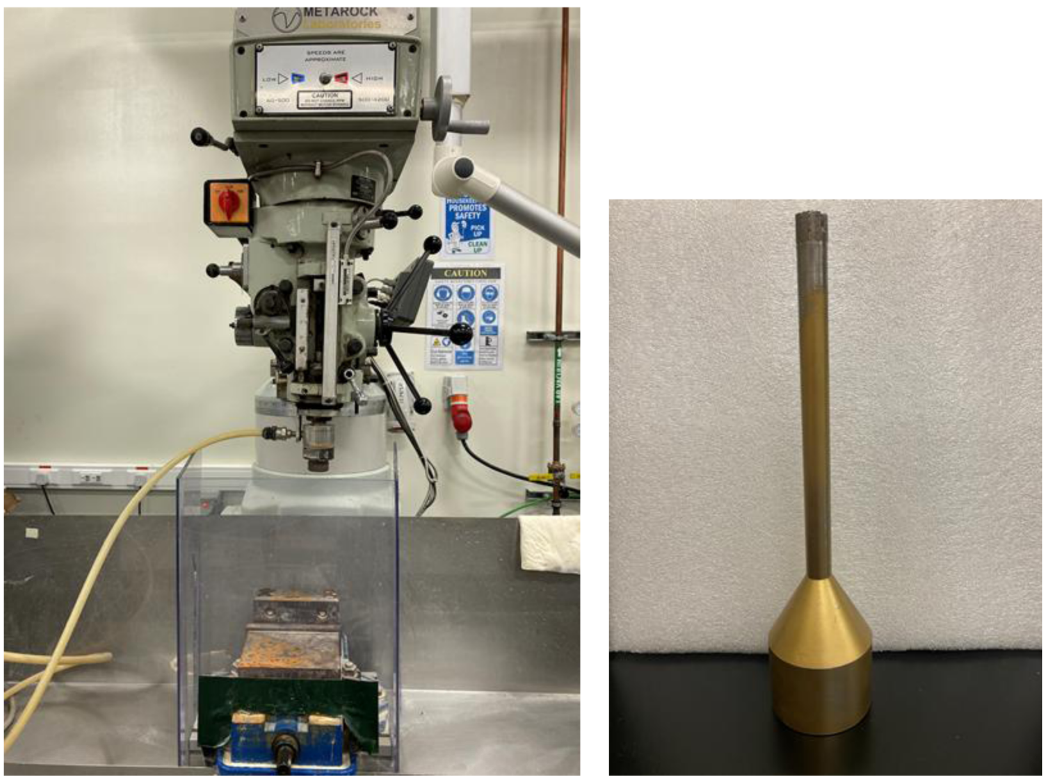

Figure 1—(left) MetaRock Universal Turret Milling Machine. (right) The drilling bit is used to obtain the micro plug. The drill bit has an internal diameter of ¼ inch. The drilling rig operates at different rpm. For the Idaho Gray sandstone, 900 rpm was used.

Experimental setup

A high-pressure, high-temperature flow cell was designed and fabricated to study the effect of different injection flow rates on the CO_2 capillary trapping. The flow cell holds the micro-core plug at high temperature and pressure while performing CT scanning. It comprises of two end caps, injection ports, a Viton sleeve, a thermocouple, and a PEEK hollow cylinder. The diameter of the flow cell is small (10 mm) to capture high-resolution images. PEEK material is X-ray transparent. Figure 2 shows a sketch of the HPHT flow cell. The flow cell was placed on the CT scan stage and connected to the flow lines and the pumps in the case of the dynamic scan. We performed time-lapse scanning to avoid complications with tubing placement during CT system rotation. The proposed schematic for the in-situ imaging is shown in Figure 2. Three ISCO pumps were used for confining, injection, and receiving. However, in the late experiments, we disconnected the receiving pump and allowed the outlet to be open to atmospheric pressure. We used several back pressure regulators to hold the CO_2 gas inside the cell, but they could not help.

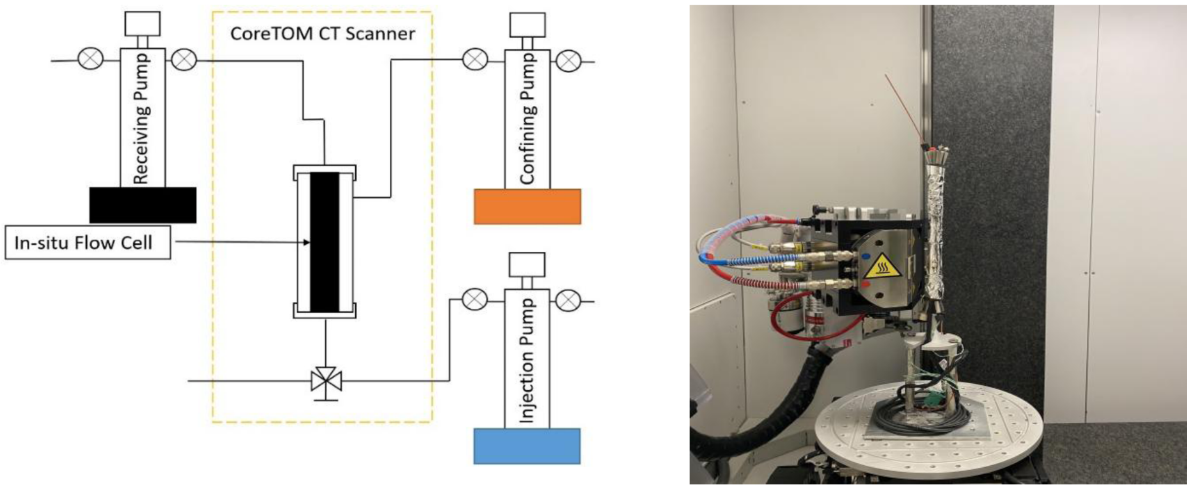

Figure 2—Schematic of the in-situ imaging experiment. The in-situ flow cell. The flow cell is mounted on the micro-CT stage and set close to the X-ray source to obtain higher resolution (Voxel size = 12 µm).

CT scan CO_2 trapping experiment. In this experiment, we assessed the influence of water injection rates on the CO_2 capillary trapping on the Idaho Gray sandstone. Figure 2 shows the flow cell setup inside the CT-scan compartment. During the CT scan experiment, a dry scan was performed to understand the initial condition of the studied rock and was used to determine the porosity and extract the pore network model. Next, the rock is initially flooded with the wetting phase, 5 wt. % CsCl solution with sufficient pore volumes to ensure full saturation. We then took another scan of the saturated rock. After that, the desaturation process began by injecting the non-wetting phase, CO_2 gas, to reach the desired residual wetting saturation. Finally, we examined the effect of different water injection rates (100, 50, and 20 µL/min) on the quantity of trapped CO_2 inside the rock. We also studied the effect of the confining pressure on CO_2 gas exsolution, and monitored the change in the CO_2 gas volume over time using CT scan imaging.

Image Processing and In-situ Contact Angle. We use the scientific image processing software package Avizo (Thermo Fisher Scientific 2022) to process the CT scan images. CT scan images were cropped, filtered, in preparation for the segmentation process. Segmented CT images estimated the plugs' initial $CO_2$and water saturation. Wetting and nonwetting phases were labeled and quantified by voxel counts. The change in CO_2 saturation is then computed. We also perform volume fraction distribution and equivalent diameter measurements to assess the behavior of CO_2 volume with time.

The change in CO_2 saturation is computed by measuring the change in CO_2 voxel counts before and after the spontaneous imbibition. We also obtain the oil volume fraction along the length of the micro plug. The CO_2 micro-occurrences distribution is studied based on the equivalent diameter D_{eq} formula (built-in Avizo software):

$$D_{eq} = \sqrt{\frac{6.V_{3D}}{\pi}} \tag{1}$$

Where V_{3D} is the computed 3D volume of each separated pore

Results and Discussion

CO_2 Capillary trapping at different rates

As discussed earlier, we aim to study how much CO_2 gas can be trapped at different injection rates during the imbibition process. Three injection rates were investigated: 0.02, 0.05, and 0.1 ml/min. Initially, we used backpressure to ensure we kept some CO_2 gas inside the system during the imbibition phase; however, no trapping could be observed after several trials. Backpressure was set to study the injection rate performance. We used the first imbibition result for each injection rate for comparison.

Figure 3 shows examples of the raw and segmented images in addition to the extracted CO_2 volume. We can observe the significant reduction in free CO_2 gas amount before and after the imbibition, leading to trapped CO_2 gas within the rock. We can also observe several large-volume CO_2 gas ganglia, which indicate preferred dissolution areas. Table 1 summarizes the volume fraction changes of the three labeled phases: grain, water, and CO_2(gas phase), during the lowinjection rate 0.02 ml/min. The amount of capillary trapped CO_2 gas is found to be ~15%. Equivalent diameters analysis in Figure 4 shows a global reduction in the CO_2 gas ganglia diameters before and after the imbibition. For this imbibition case, we also have used the labeled images to compute the contact angle. Figure 5 shows the in-situ contact angle measurement distribution for this case with an average of 45°, indicating the water-wet system expected for the used sandstone sample. The average porosity of the Idaho Gray sandstone dray sample is 0.30, and some heterogeneity is observed in the volume fraction analysis, as in Figure 6. The dry scan result is used for comparison later.

Table 1—Change in volume fraction of CO_2 during the 0.020 ml/min imbibition

Volume Fraction			
Scan	**CO_2**	**Water**	**Grains**
Drainage	0.114	0.106	0.780
Imbibition first	0.017	0.185	0.798

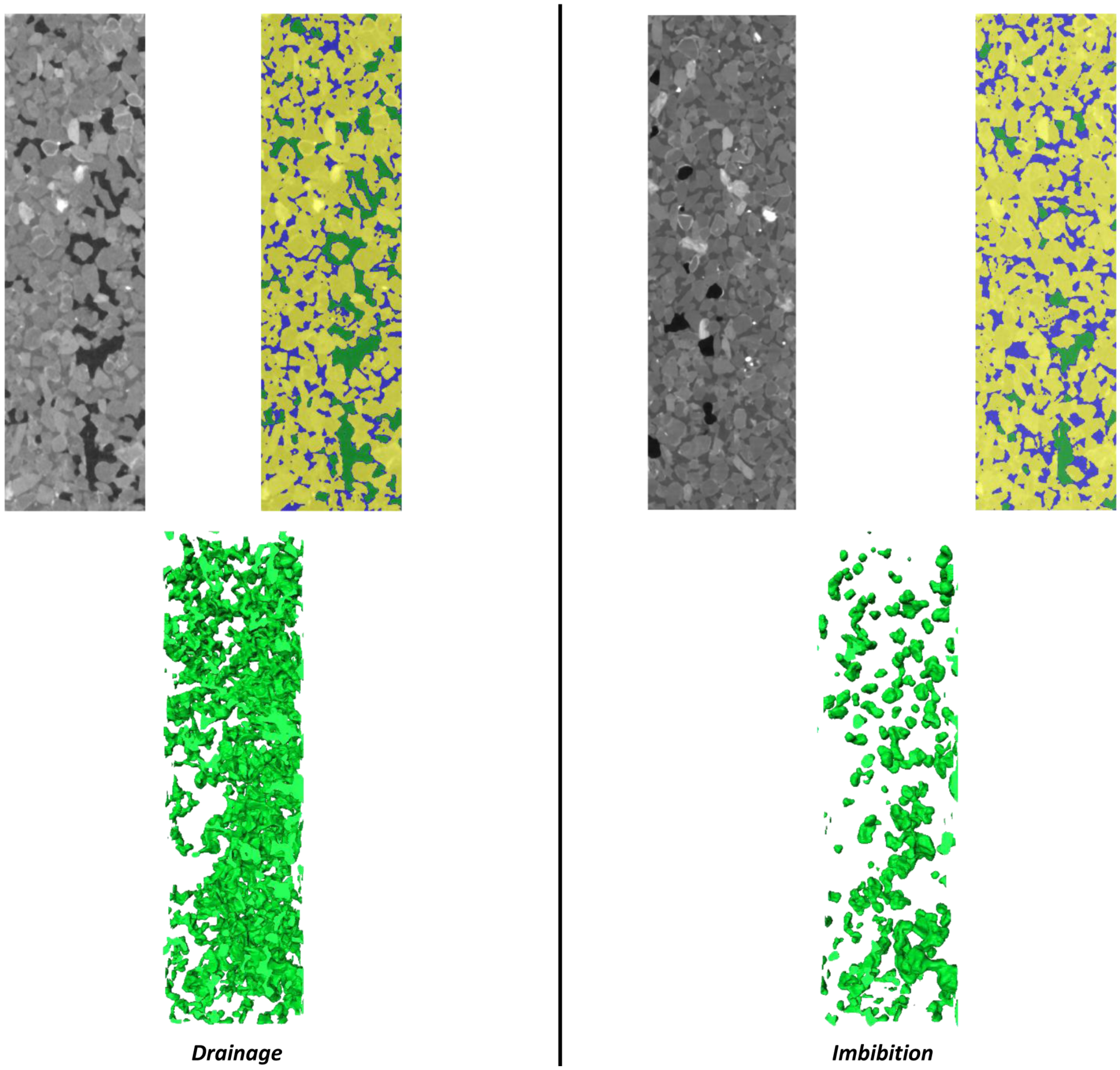

Figure 3—Initial and last scan results of CO_2 phase during the 0.020 ml/min imbibition.

Imbibition first	0.017	0.185	0.798

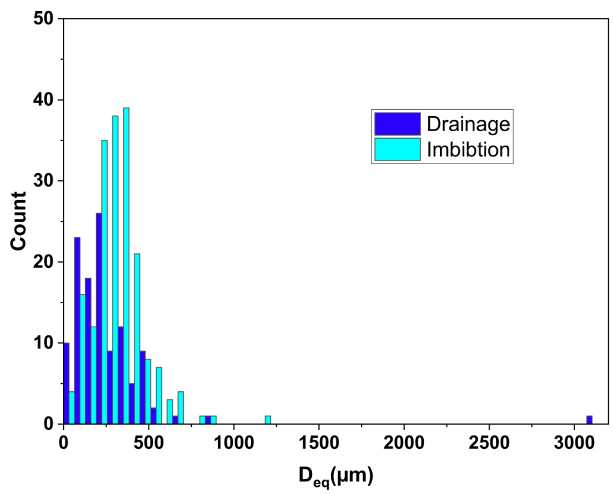

Figure 4—Change in equivalent diameters between drainage and imbibition process for 0.02 ml/min imbibition.

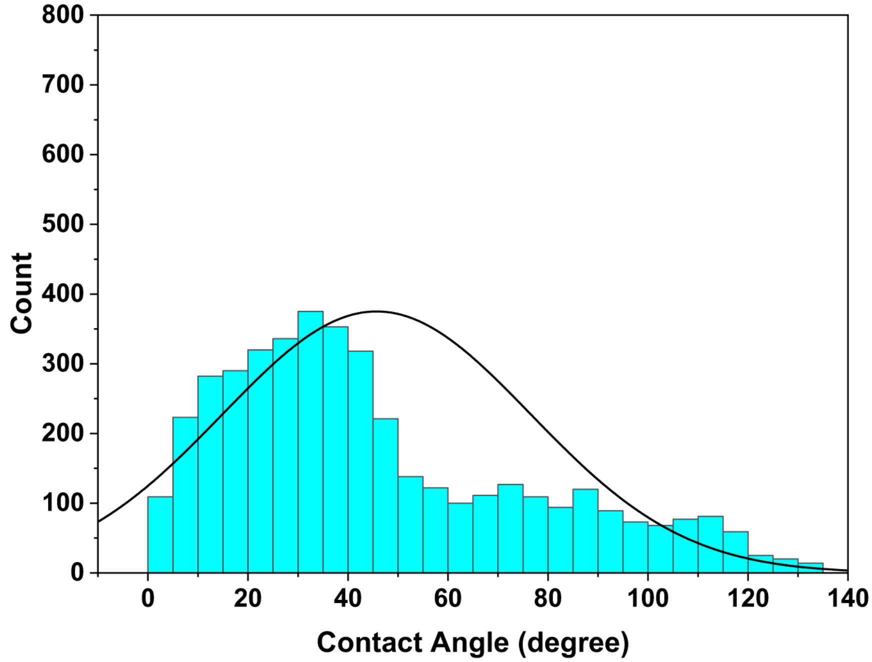

Figure 5—In-situ contact angle analysis of the 0.02 ml/min imbibition case. The average contact angle is around 45°, indicating a water-wet system.

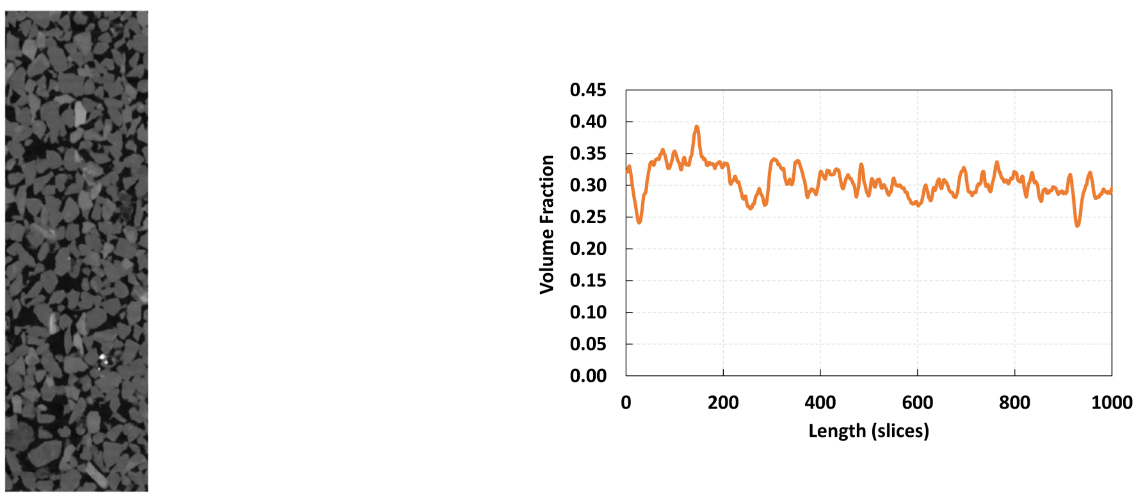

Figure 6—Dray rock scan and porosity measurement along the length of the Idaho Gray sample. Porosity is computed to be 0.30 with some heterogeneity that appears at the bottom of the sample (0-100 slices).

For the 0.05 ml/min imbibition case, Figure 7 summarizes the initial and final results, including the raw image, segmented image, and the extracted CO_2 volume. Several CO_2 gas ganglia can be identified and considered as preferred dissolution areas. Table 2 summarizes the volume fraction changes of the three labeled phases: grain, water, and CO_2 gas. The amount of capillary trapped CO_2 gas is found to be ~34%, significantly higher than the lower injection rate. The volume fraction analysis in Figure 8 indicates that CO_2 capillary trapping occurs with the same magnitude in all slices. Equivalent diameters analysis in Figure 9 shows a global reduction in the CO_2 gas ganglia diameters before and after the imbibition, indicating the trapping of the CO_2gas molecules.

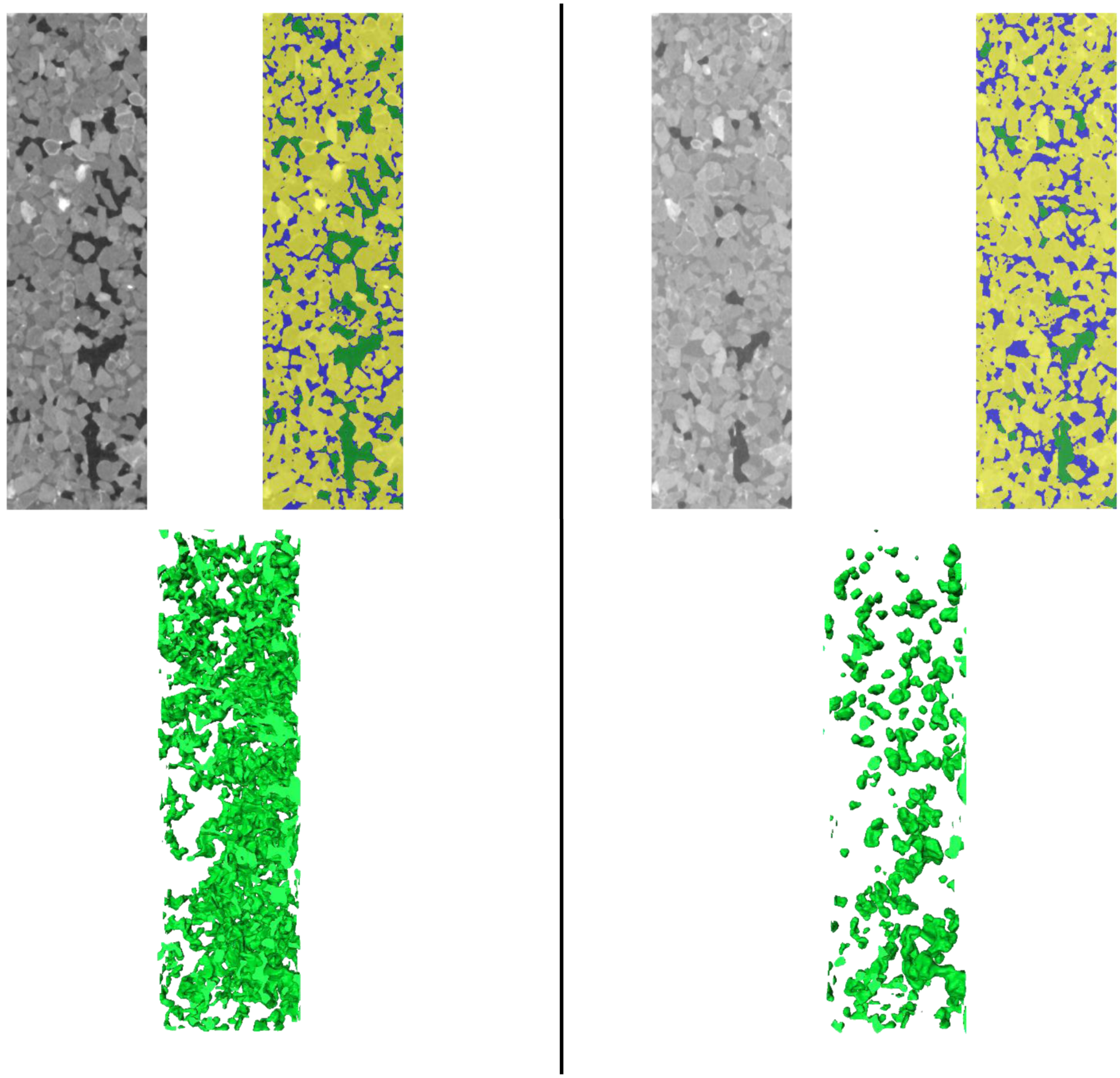

Figure 7—Initial and last scan results of CO_2 during the 0.05 ml/min imbibition.

Table 2—Change in volume fraction of CO_2 during the 0.05 ml/min imbibition

Volume Fraction			
Scan	**CO_2**	**Water**	**Grains**
Drainage	0.106	0.152	0.74
Imbibition	0.036	0.194	0.77

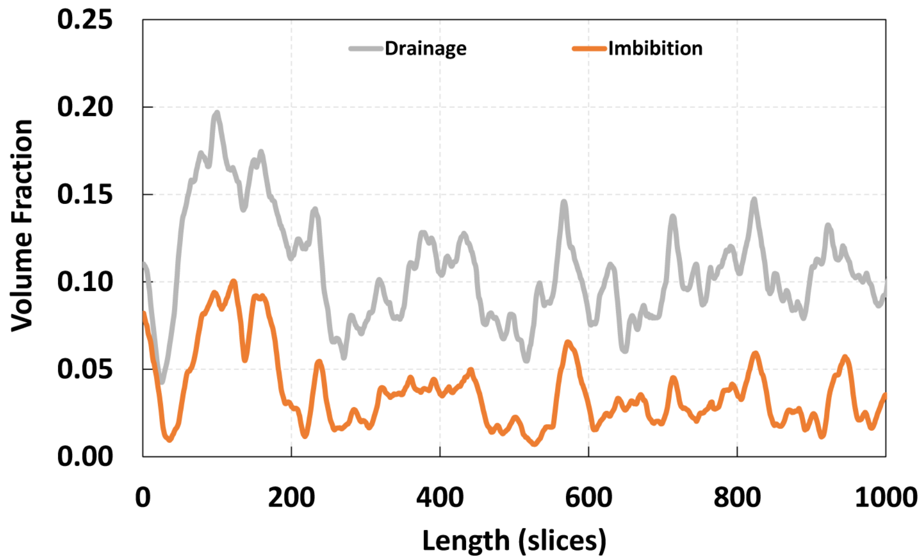

Figure 8—Volume fraction change in initial and final stages. The final stage shows a discontinuity in the CO_2 plume at different areas at 0.05 ml/min.

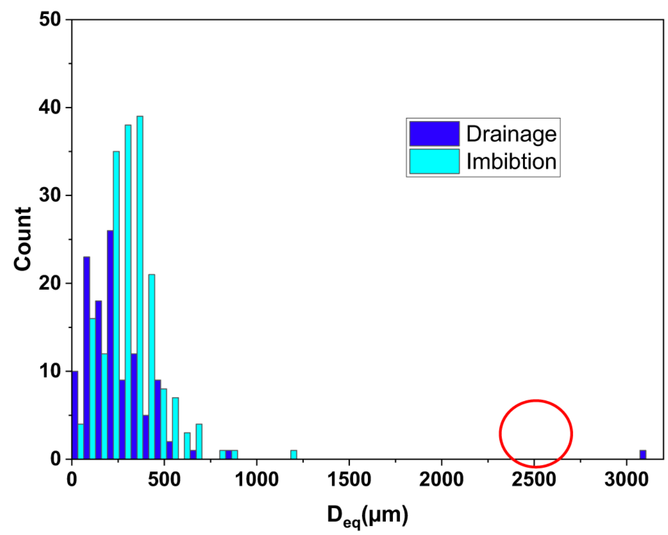

Figure 9—Change in equivalent diameters between drainage and imbibition process. In the drainage, one big, connected volume that creates a network through the sample is observed with an equivalent diameter greater than 3.0 mm (indicated with a red circle).

For the highest imbibition rate, 0.100 ml/min, the amount of capillary trapped CO_2 Gas is calculated to be 28%. Figure 10 (top) shows 2D slices of the raw images before and after the drainage, and the. Figure 10 (bottom) shows the additional analysis we did to assess the volume of the CO_2 ganglia after the imbibition. The different colors represent different ganglia. We can clearly observe how the imbibition affected the volume and shape of the ganglia, making them trapped, disconnected, and occupied rounded shapes. We still can observe some higher diameters ganglia, which can lead to some solubility of CO_2 with time.

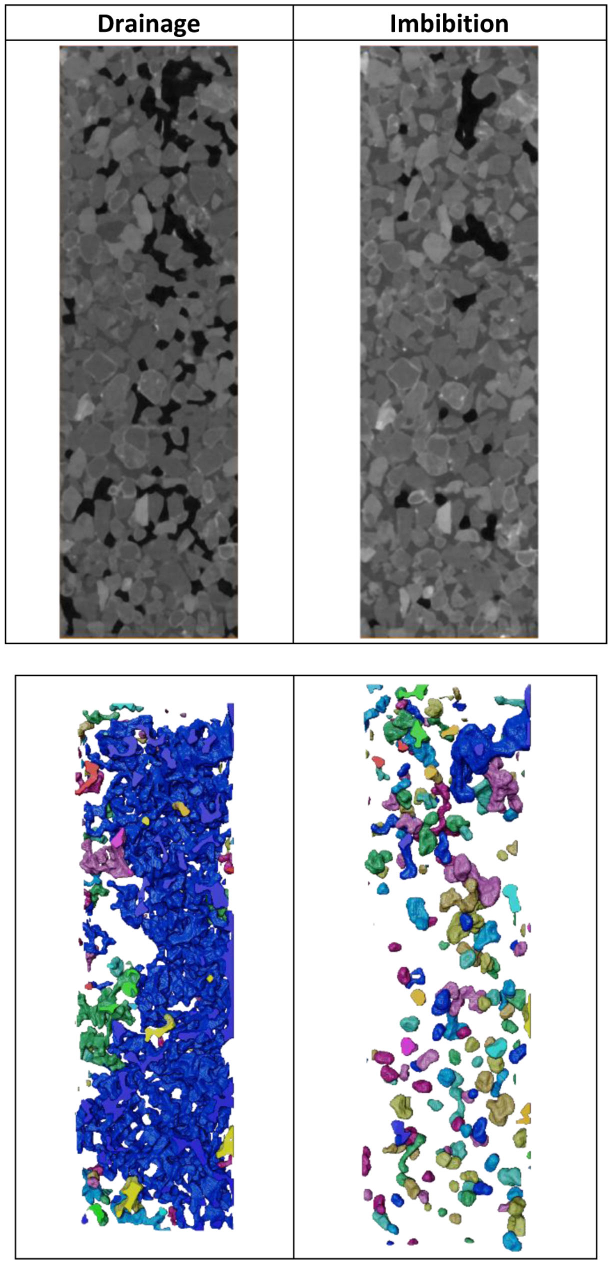

Figure 10—(top) Initial and last scan results of CO_2 during the 0.1 ml/min imbibition (bottom) separated ganglia based on the equivalent diameters. The size and shape of the initial CO_2 volume are affected by the imbibition process.

Figure 11 summarizes the results of the three injection rates. Since the capillary trapping depends on the capillarity effects, the hypothesis indicates that at a lower capillary number, higher nonwetting phase, CO_2 in this case, can be trapped. However, here, we can observe that the lower injection rate did not lead to the highest trapped CO_2.

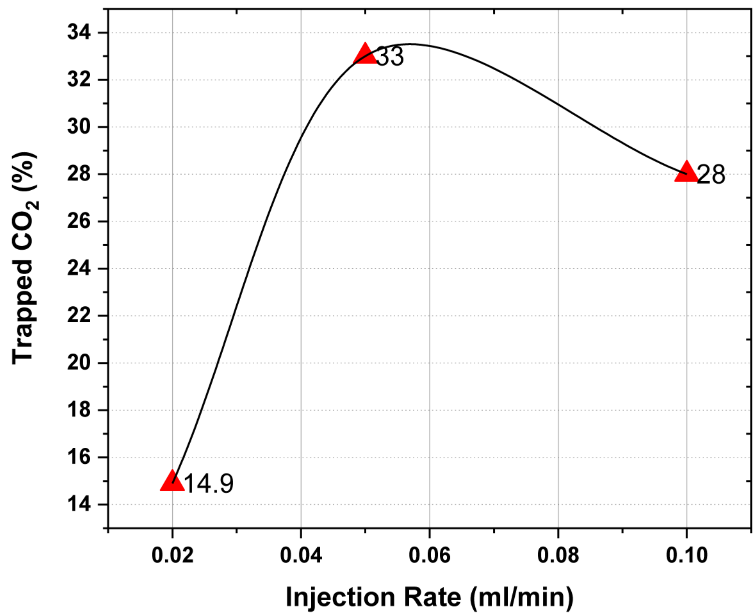

Figure 11—Change in trapped CO_2 with the change in the imbibition rates.

At a low flow rate (0.02 ml/min), the capillary number would be small, indicating that surface tension forces dominate. This could lead to a high degree of capillary trapping initially, but if the Bond number is also high (indicating that gravity forces are significant), the CO_2 might be pushed out due to the buoyance, resulting in lower overall trapping.

At a medium flow rate (0.05 ml/min), the Capillary number increases, implying that viscous forces become slightly important. However, at the same time, the bond number becomes slightly lower, leading to less buoyancy effect. This might lead to a relatively balanced case between gravity, viscosity, and capillarity forces, leading to the highest trapped CO_2.

The capillary number is even larger at a high flow rate (0.1 ml/min). However, if the viscous forces become too dominant, it may reduce capillary trapping due to less time for CO_2 to interact with the pore walls and be trapped. This could explain the intermediate level of trapping observed at this flow rate.

CO_2 solubility at different times

In this work, we monitored the change in CO_2 volume over time injection. We monitored the 0.02 and 0.1 ml/min injection experiments. Both were monitored for almost 12 hours. Two scans were performed for the low injection one, and a time-lapse scanning of 6 scans was done for the high injection rate experiment.

Figure 12 compares raw, segmented, and CO_2 extracted volumes for the 0.02 ml/min injection rate initially and after 12 hours. The volume fraction of the CO_2 is reduced from 0.017 to 0.011, representing a 35.3% reduction. The equivalent diameters distribution indicates that smaller CO_2 ganglia are formed, as shown in Figure 13. CO_2 volume fraction analysis, Figure 14 revealed that the reduction in the CO_2 volume happens throughout the rock length with no preferred dissolution areas.

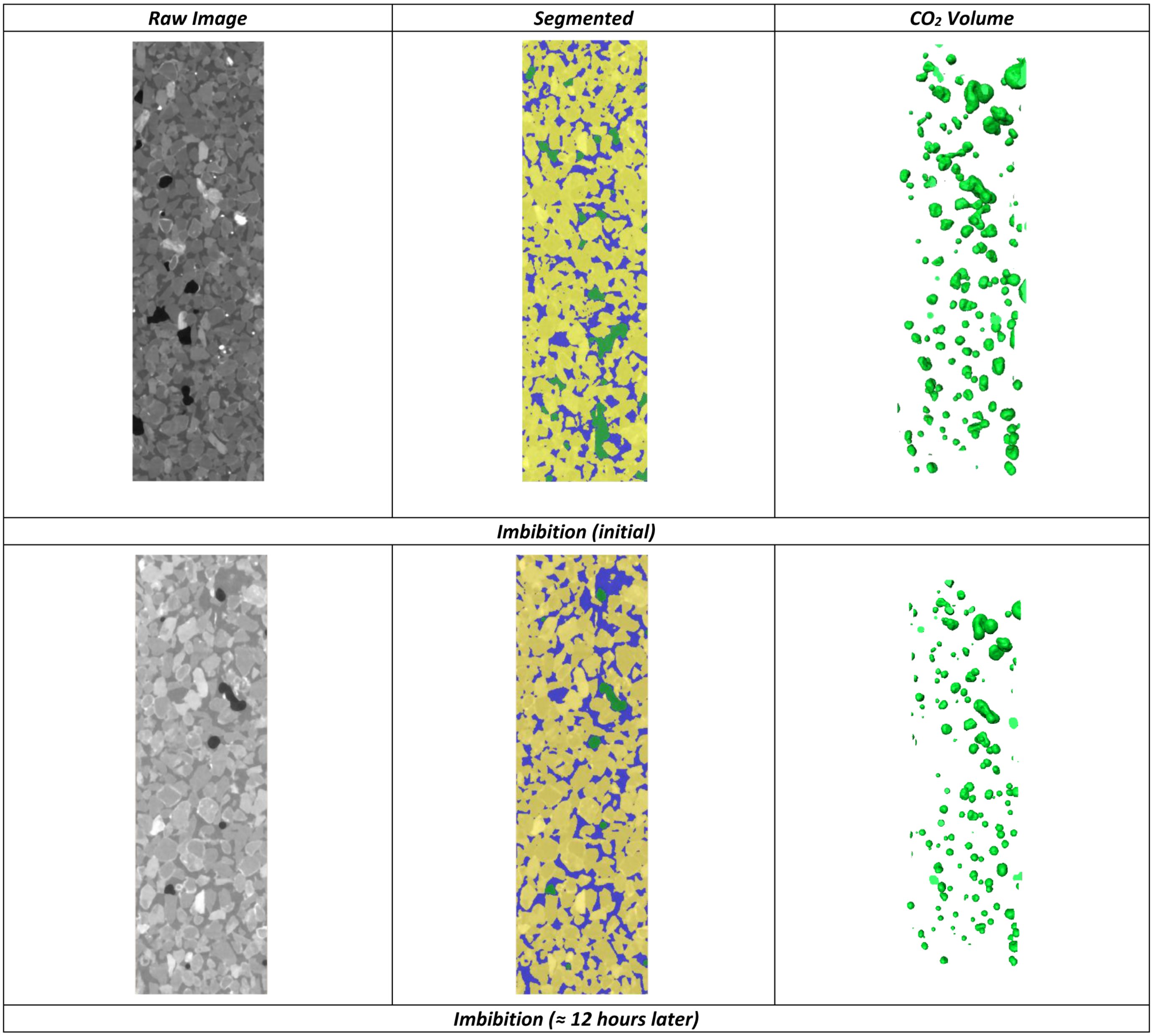

Figure 12—Change in CO_2 volume with time at 0.02 ml/min injection rate.

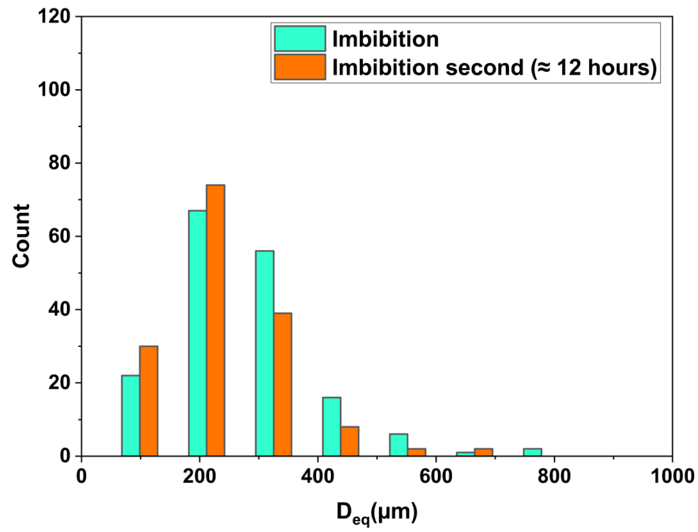

Figure 13—Change in equivalent diameters per time for the 0.02 ml/min injection rate experiment.

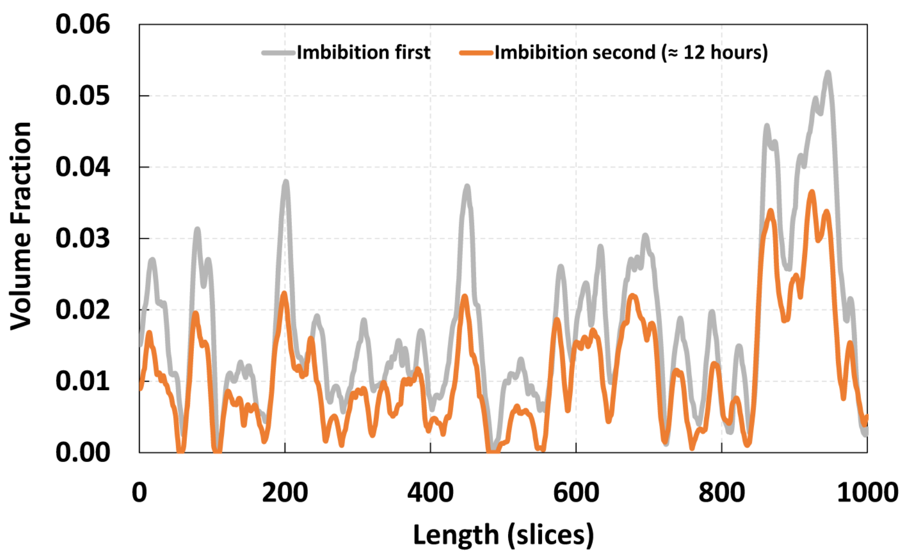

Figure 14—Volume fraction change in trapped CO_2 with time. The second time shows a decrease in the CO_2 volume, which indicates the dissolution of CO_2 with time.

On the other hand, the trapped CO_2 after the 0.1 ml/min injection was monitored with time-lapse scanning to assess the dynamics of the CO_2 solubility. Seven scans were done with 2 hours difference. Figure 15 shows the change in CO_2volume fraction with time during the seven scans. We can observe a noticeable decrease at the beginning, then an increase after some time, and then a decrease again. The slight increase in the CO_2 concentration might be attributed to two reasons: artifacts in the image processing or re-distribution of the CO_2 molecules within the rock. The overall reduction in the 12 hours is calculated to be 23.18% in volume. The volume fraction analysis did not indicate any preferred dissolution zone, as the reduction happens in almost all slices, as depicted in Figure 16.

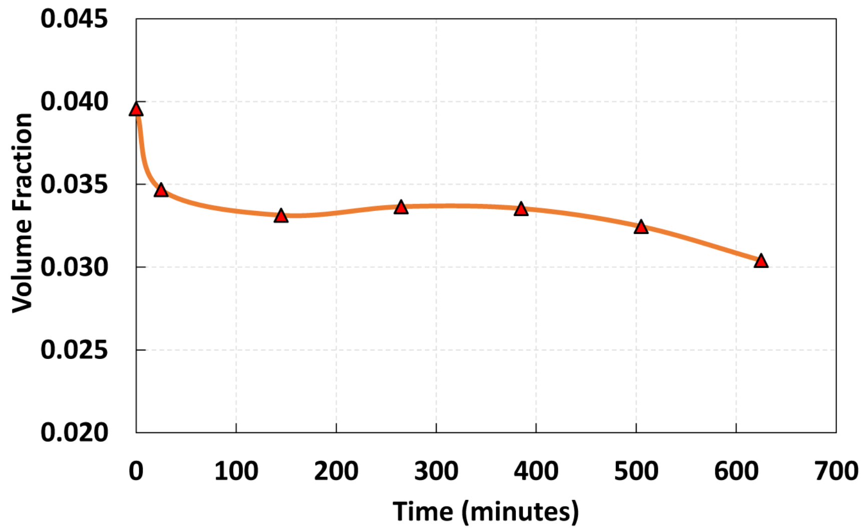

Figure 15—Change in CO_2 volume fraction with time for 0.1 ml/min injection rate.

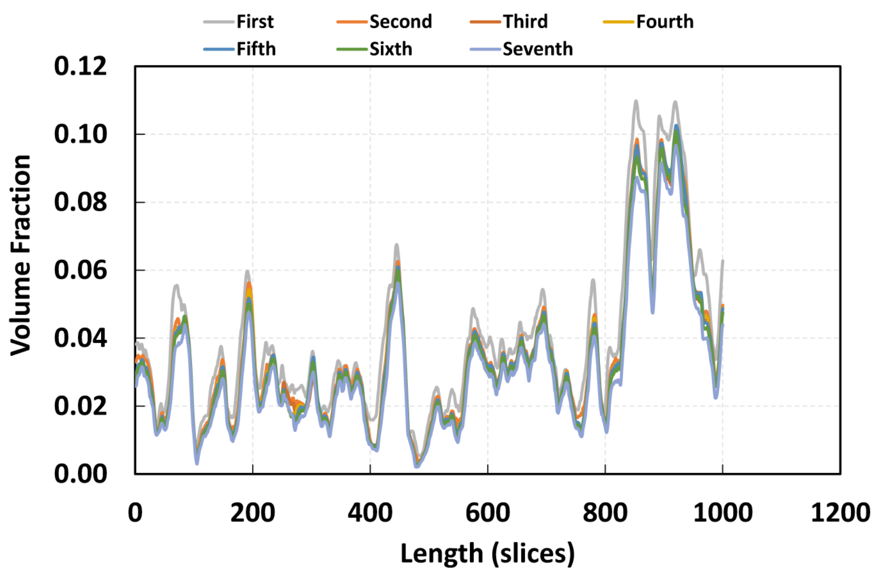

Figure 16—Volume fraction change in trapped CO_2 with time. The solubility happens in most slices throughout the rock without indicating CO_2 leakages outside the flow cell.

Effect of change in pressure on the CO_2 solubility

In this research, we delve into the complexities of CO_2 exsolution dynamics from aqueous solutions due to variations in pressure. Such a process parallels the phenomena observed in subsurface reservoirs, specifically during the injection of CO_2-laden water into basalt formations.

In the context of CO_2 sequestration, the injection of carbonated water into basaltic formations induces an initial increase in formation pressure. Following the cessation of the injection process, a subsequent decrease in formation pressure is anticipated. These pressure fluctuations, both escalation and reduction, can significantly influence the solubility of CO_2 within the aqueous solution. Understanding these pressure-dependent CO_2 exsolution dynamics is crucial for predicting the behavior of CO_2 in carbon capture and storage applications, and for optimizing these processes to ensure maximum CO_2 sequestration and minimal environmental impact.

In our experimental setup, we initially saturated the rock sample with water, and proceeded with a CO_2 drainage process until reaching a predetermined saturation level. Subsequently, we modulated the initial confining pressure from 150 psi down to 100 psi to scrutinize the CO_2 exsolution process.

Figure 17 depicts the change in CO_2 volume fraction with time. The volume of CO_2 almost doubled during the 24 hours from 0.008 to more than 0.014. The change rate is seen faster in the first 5 to 6 hours, with almost less than a 10% increase during the last 14 hours. Figure 18 depicts the evolution of CO_2 ganglia over time. As time progresses, the CO_2 ganglia show a trend of connecting and increasing volume. This growth and connectivity of the CO_2 ganglia might be attributed to the Ostwald ripening phenomenon. Equivalent diameters analysis, as in Figure 19, also confirms the change in CO_2 ganglia as larger ganglia diameters formed later.

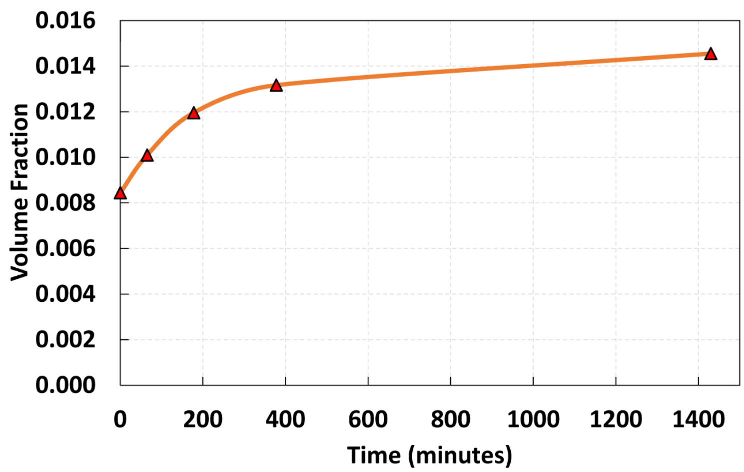

Figure 17—Change in CO_2 gas volume with time after changing confining pressure from 150 to 100 psi.

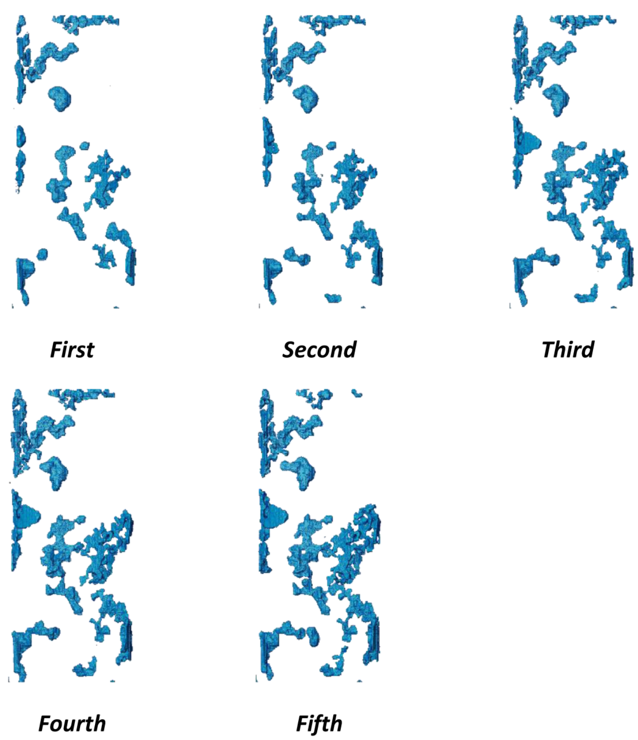

Figure 18—Change in CO_2 volume with time as a result of the reduction in confining pressure and release of the dissolved CO_2 bubbles.

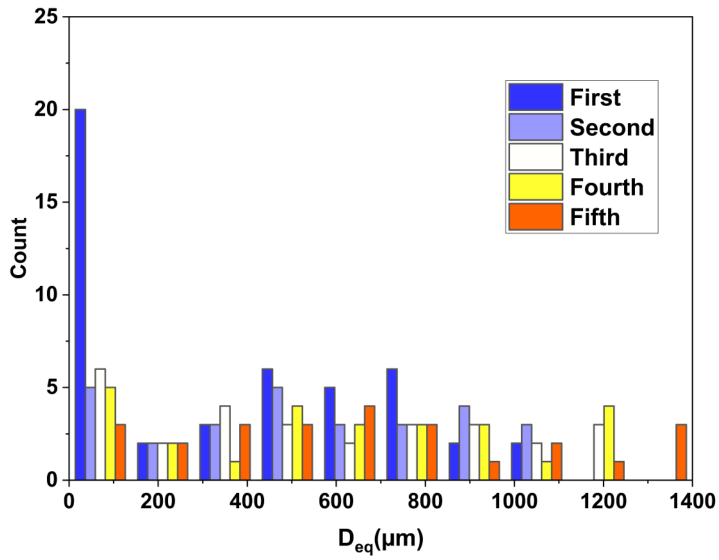

Figure 19—Distribution of equivalent diameters with time. Larger diameters are formed as CO_2 bubbles connect with time.

To further explore the effects of varying pressures on carbon dioxide (CO_2) solubility, another experimental study was conducted. This investigation employed a broader range of pressures, spanning from 50 to 300 psi. Figure 20 and Figure 21 summarizes the change in CO_2volume fraction at the different pressure values and time. The higher the pressure, the lower the CO_2 gas volume which indicates more solubility of CO_2.

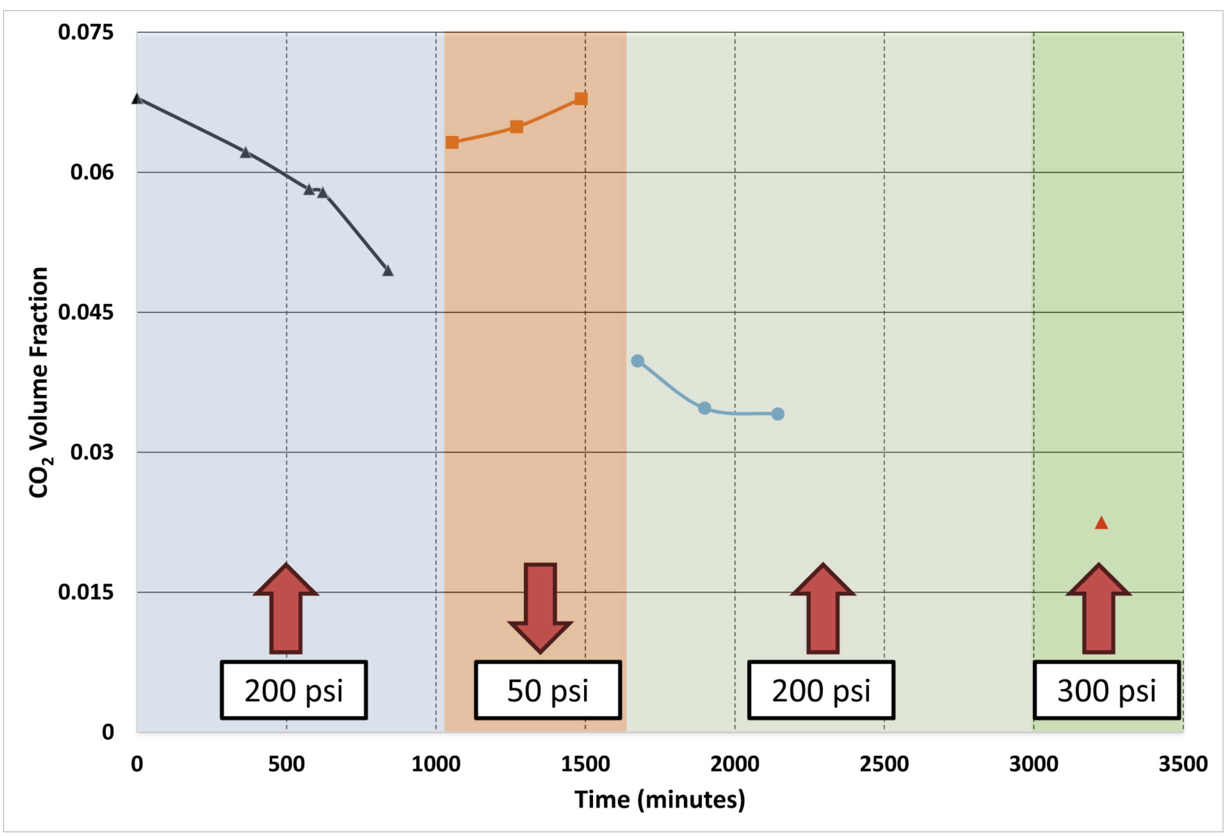

Figure 20—Change in CO_2 volume fraction as the confining pressure changes

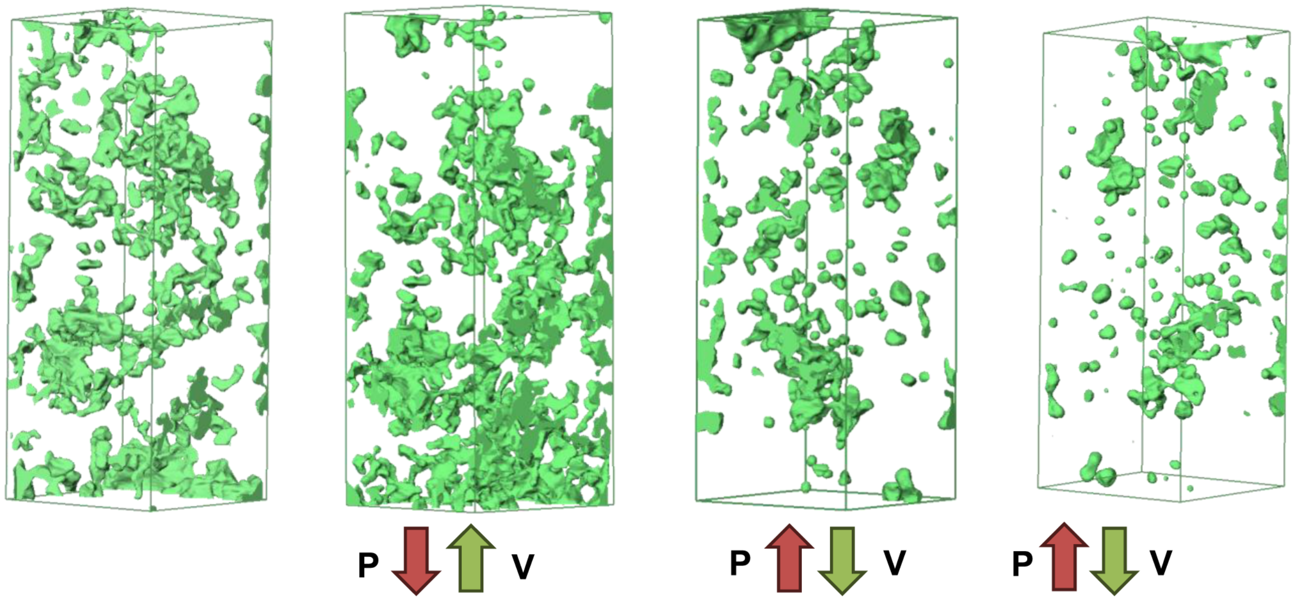

Figure 21—Change in CO_2 Volume during the change in the confining pressure

The results, demonstrate that increasing pressure leads to increased solubility of CO_2, resulting in a decrease in its volume fraction. Specifically, when the confining pressure increases, the CO_2he CO_2 volume fraction decreases correspondingly. These findings indicate that fluctuations in pressure significantly influence the solubility of CO_2, underscoring the importance of considering pressure dynamics in applications involving supercritical fluids.

Conclusion

In this work, we investigated CO_2 capillary trapping as one of the CO_2 sequestration mechanisms at the pore-scale level. While most previous studies focus on the supercritical phase of CO_2, we aim to study the effect of water imbibition injection rate on the quantity and distribution of CO_2 gas capillarity trapping. Three main forces are involved in this mechanism: viscous force due to injection force, buoyancy due to significant differences in density, and capillarity force due to interfacial tension between the phases. In-situ imaging of CO_2 trapping at different times and injection rates was performed. Quantitative and qualitative analysis for the trapped CO_2 is conducted. We also assessed the potential of CO_2 solubility trapping by monitoring the change in CO_2 volume with time.

The main findings are summarized as follows:

- Three injection rates are studied: 0.02, 0.05, and 0.1 ml/min. Each cycle includes primary imbibition by water, drainage by CO_2, and secondary imbibition by water again at the proposed rate.
- The injection rate of 0.05 ml/min provides the highest initial trapped CO_2 compared to the other two, with 33% trapped CO_2 and 14.9% and 28% for 0.02 and 0.1 ml/min injection rates, respectively.
- The gravitational force competes with the capillary at a very low imbibition rate, and CO_2 is pushed out (high bond number). In contrast, viscous forces dominate at a high injection rate, and water displaces CO_2 gas (high capillary number). In these two cases, the capillary trapping is minimal compared to the medium imbibition rate, where the capillary is at its maximum.
- The rate of CO_2 solubility increases with the increase in capillary trapped CO_2 gas quantity as there is a higher water-CO_2 surface area of interaction.

- Pore-scale imaging reveals that CO_2 gas during dissolution gets smaller and rounded shape, while in the release process, it gets larger and connected.

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